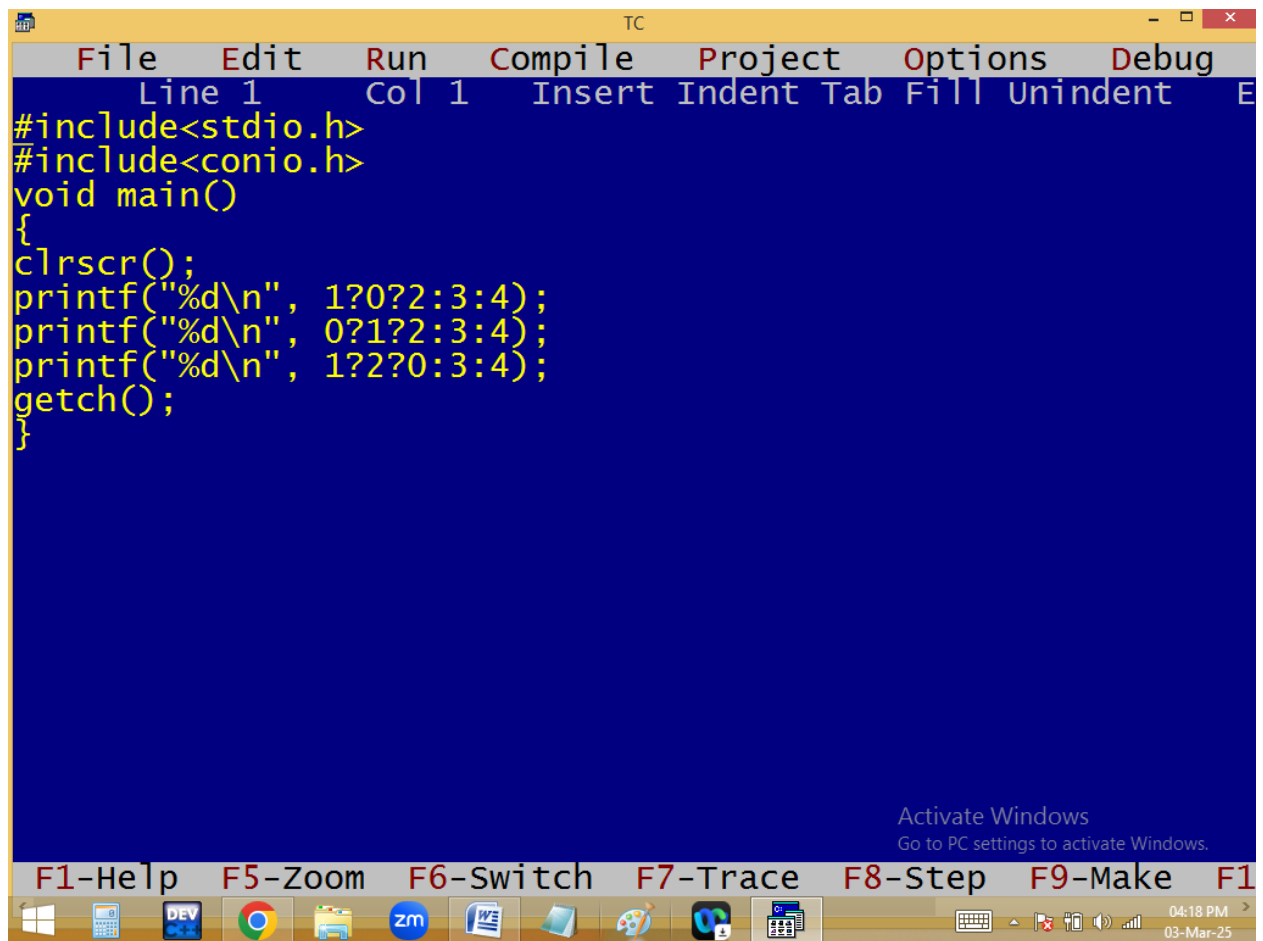


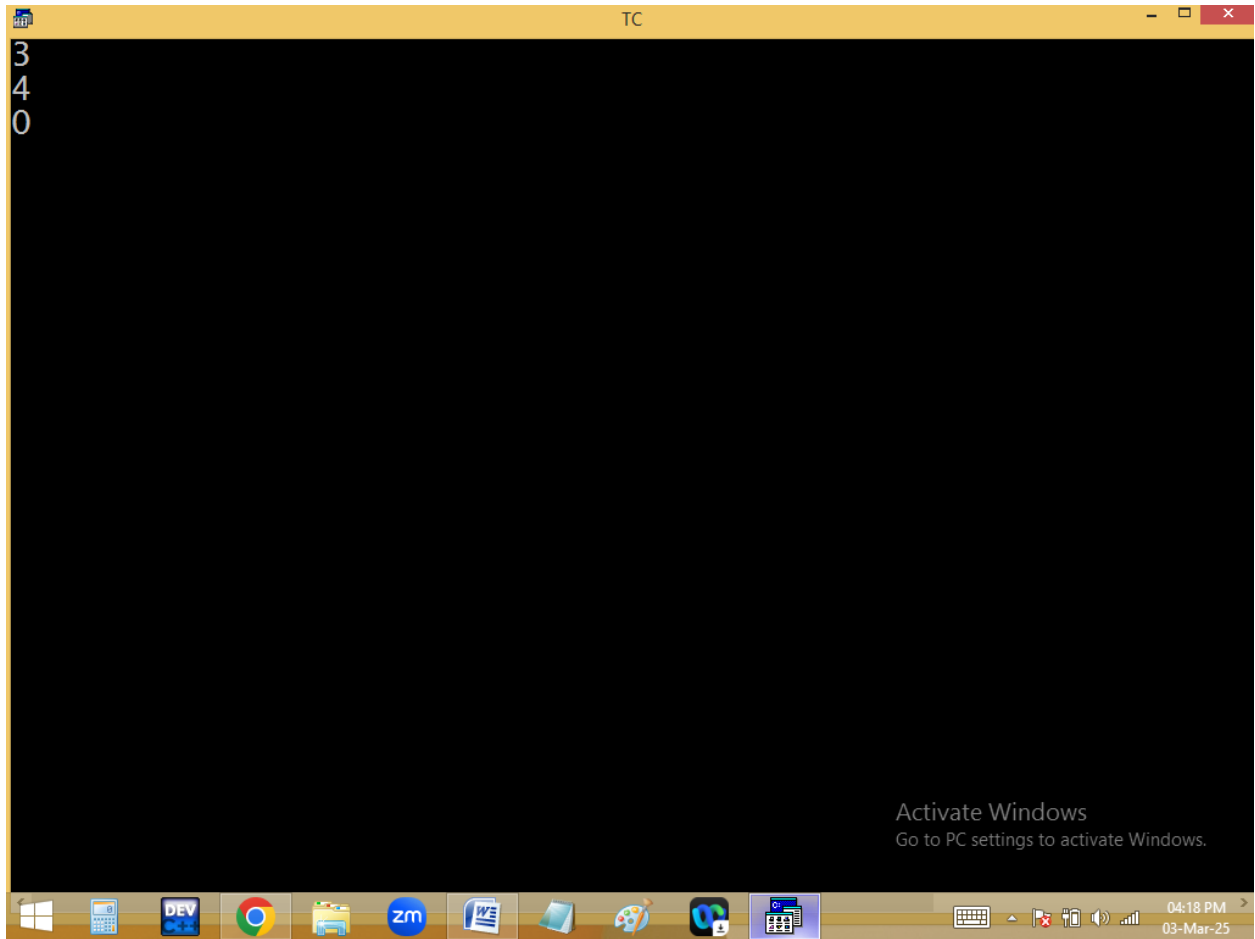


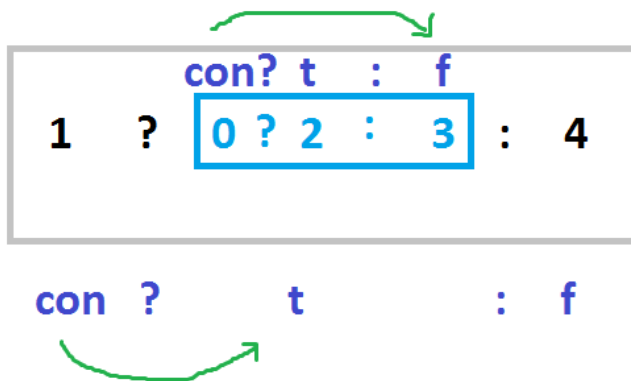
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, there is a status bar showing "Line 1", "Col 1", and "Insert Indent Tab Fill Unindent E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 1?0?2:3:4);
printf("%d\n", 0?1?2:3:4);
printf("%d\n", 1?2?0:3:4);
getch();
}
```

At the bottom of the window, there is a taskbar with several icons: Windows Start button, Task View, File Explorer, Google Chrome, Zoom, Notepad, and others. The system tray on the right shows the time "04:18 PM" and the date "03-Mar-25".

Below the code, there is a footer bar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Exit".

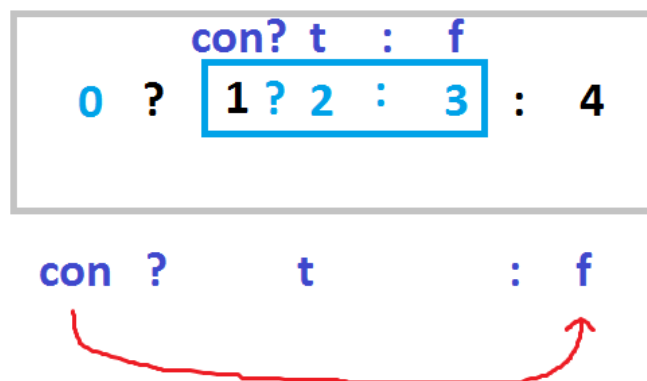




```

if ( 1 )
{
  if( 0 ) p( 2 );
  else    p( 3 ); ✓
}
else     p( 4 );

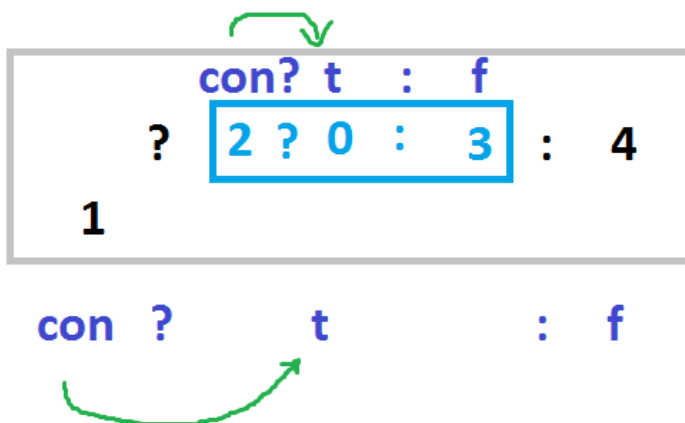
```



```

if ( 0 )
{
  if( 1 ) p( 2 );
  else    p( 3 );
}
else     p( 4 );

```



```

if ( 1 )
{
  if( 2 ) p( 0 );
  else    p( 3 );
}
else     p( 4 );

```

Switch conditional statement:

It is a selection statement.

It is used to execute one case of statements from no of cases according to the switch expression value matched with case expression value. In switch the program is jumped to matching case like the go to label.

It is similar to ladder if in working style.

Switch performance is high when compared with ladder if because of it jumps to matching case.

Syntax:

```
switch(condition / expression)  
{  
  case  constexp1:  
    statements;  
    break;  
  case  constexp2:  
    statements;  
    break;  
  case  constexpN:  
    statements;  
    break;  
  [ default: statements; ]  
}
```

Here switch, case, break, default are the keywords.

In between case and case expression / value at least one space should be provided. **Otherwise it will become a label.**

case expression/value should be a **constant integer/char value**. i.e. float / string not allowed.

One case contains one expression only.

case expression doesn't contain any separators like , . etc.

case expression should be end with : (colon)

Each case should be separated with break keyword. Otherwise remaining cases also executed.

Duplicate cases not allowed.

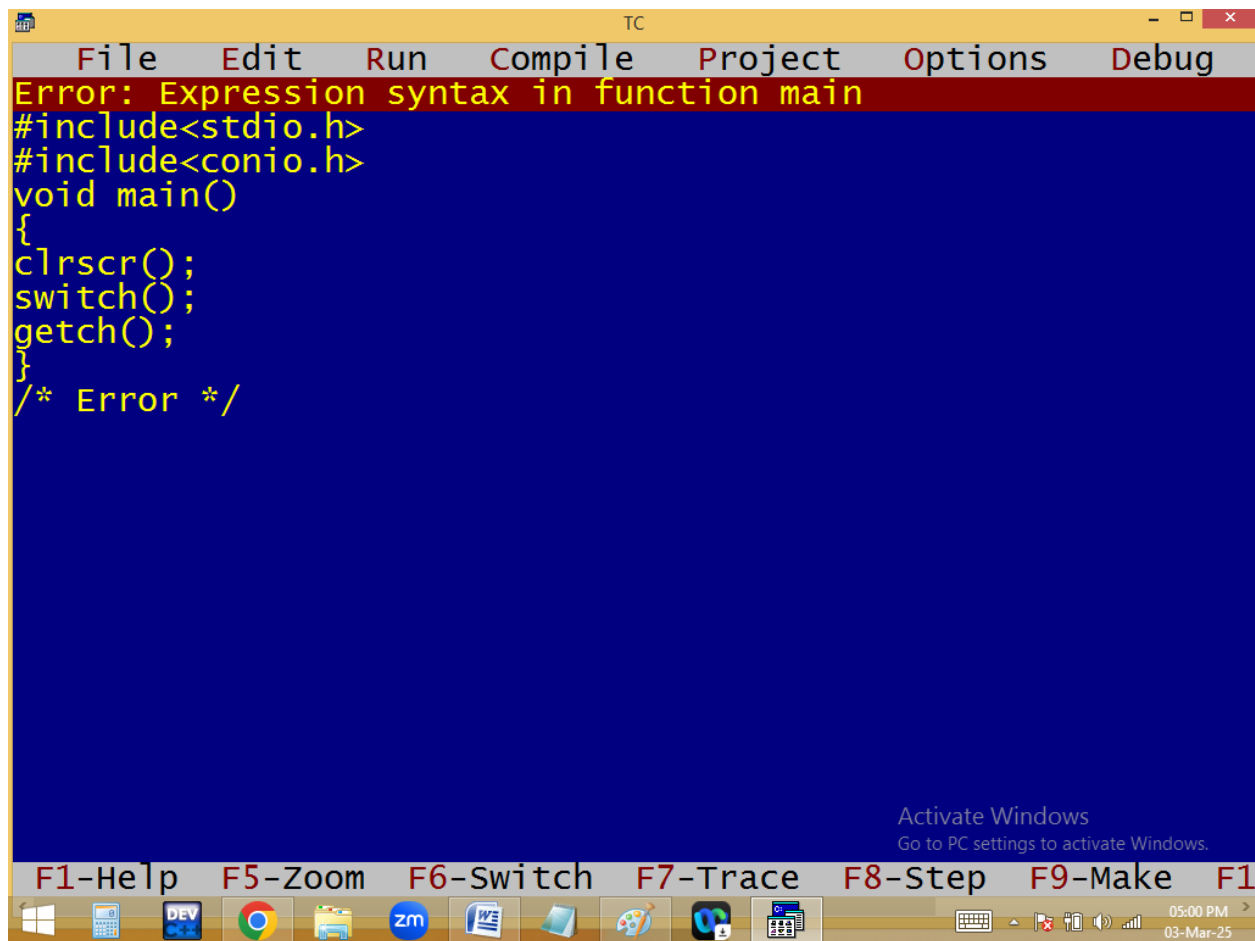
default is similar to the else and all cases are failed then default statements are executed. Default is optional and we can declare it anywhere in our switch.

Outside case expressions not considered in switch.

1. Case value should be a const int.
2. Variables not allowed in case.
3. Float / string not allowed.
4. Duplicate cases not allowed.
5. Case and case value have to separated with at least one space.
6. If space not provided then it is converted into a label.

- 7. Every case should be end with :**
- 8. Break is optional and when it is not there, remaining case also executed until the next break is occurred.**
- 9. Case doesn't allows range of values.**
- 10. Case doesn't allows , . like separators,**
- 11. Default is optional and working similar to else. We can declare default anywhere.**
- 12. Switch, case, break and default are the keywords.**

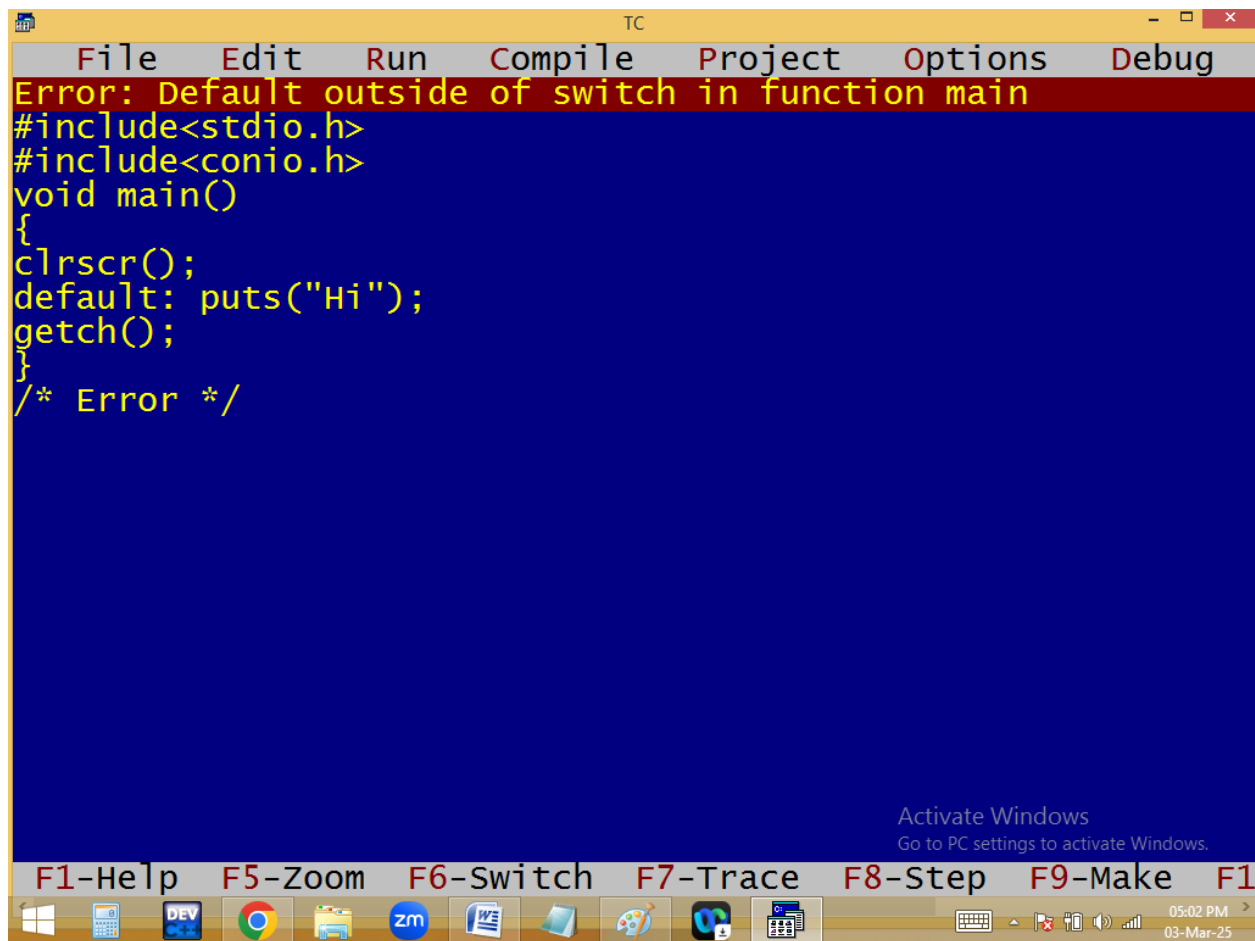
1. case value should be a const int.
2. float / strings not allowed.
3. var not allowed.
4. dup not allowed.
5. break is optional
6. default optional.
7. range of values not
8. , . not allowed.
9. outside case exp not considered.
10. switch / case / break / default
11. case and case value should be separated with min 1 space.
12. if no space provided it is converted into a label.
13. every case should have to end with :



The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the editor area states "Error: Expression syntax in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch();
getch();
}
/* Error */
```

At the bottom of the IDE window, there is a status bar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Google Chrome, Zoom, Word, and several other applications. The system tray on the right shows the time as "05:00 PM" and the date as "03-Mar-25". A watermark "Activate Windows" is visible in the bottom right corner of the desktop area.



The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code editor states: "Error: Default outside of switch in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
default: puts("Hi");
getch();
}
/* Error */
```

At the bottom of the IDE window, there is a status bar with the text "Activate Windows" and "Go to PC settings to activate Windows." Below this, a row of function key shortcuts is displayed: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Exit". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, Task View, File Explorer, and several other applications. The system clock in the bottom right corner shows "05:02 PM" and "03-Mar-25".

TC

File Edit Run Compile Project Options Debug

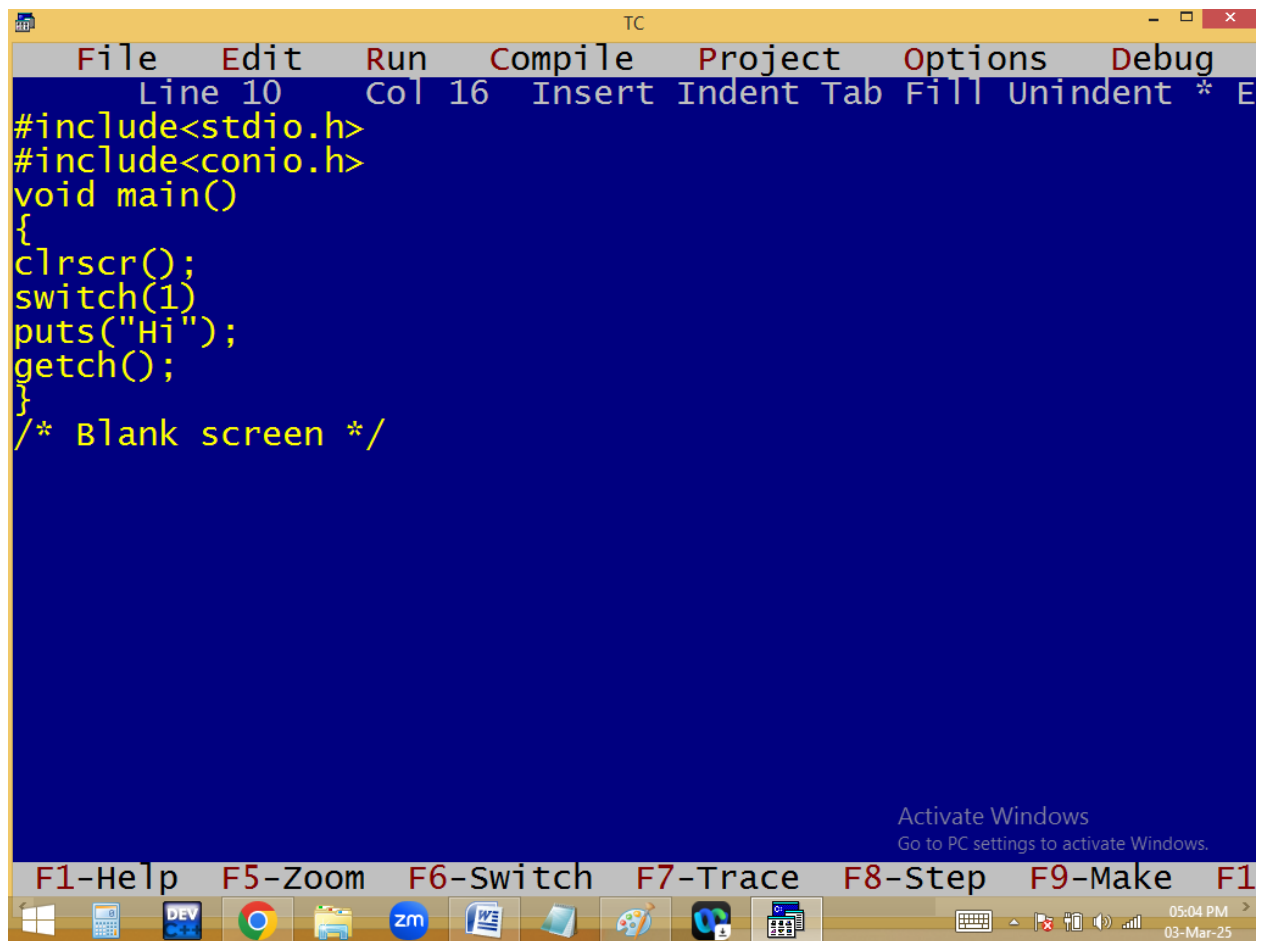
Error: Case outside of switch in function main

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
case 1: puts("Hi");
getch();
}
/* Error */
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

05:02 PM
03-Mar-25

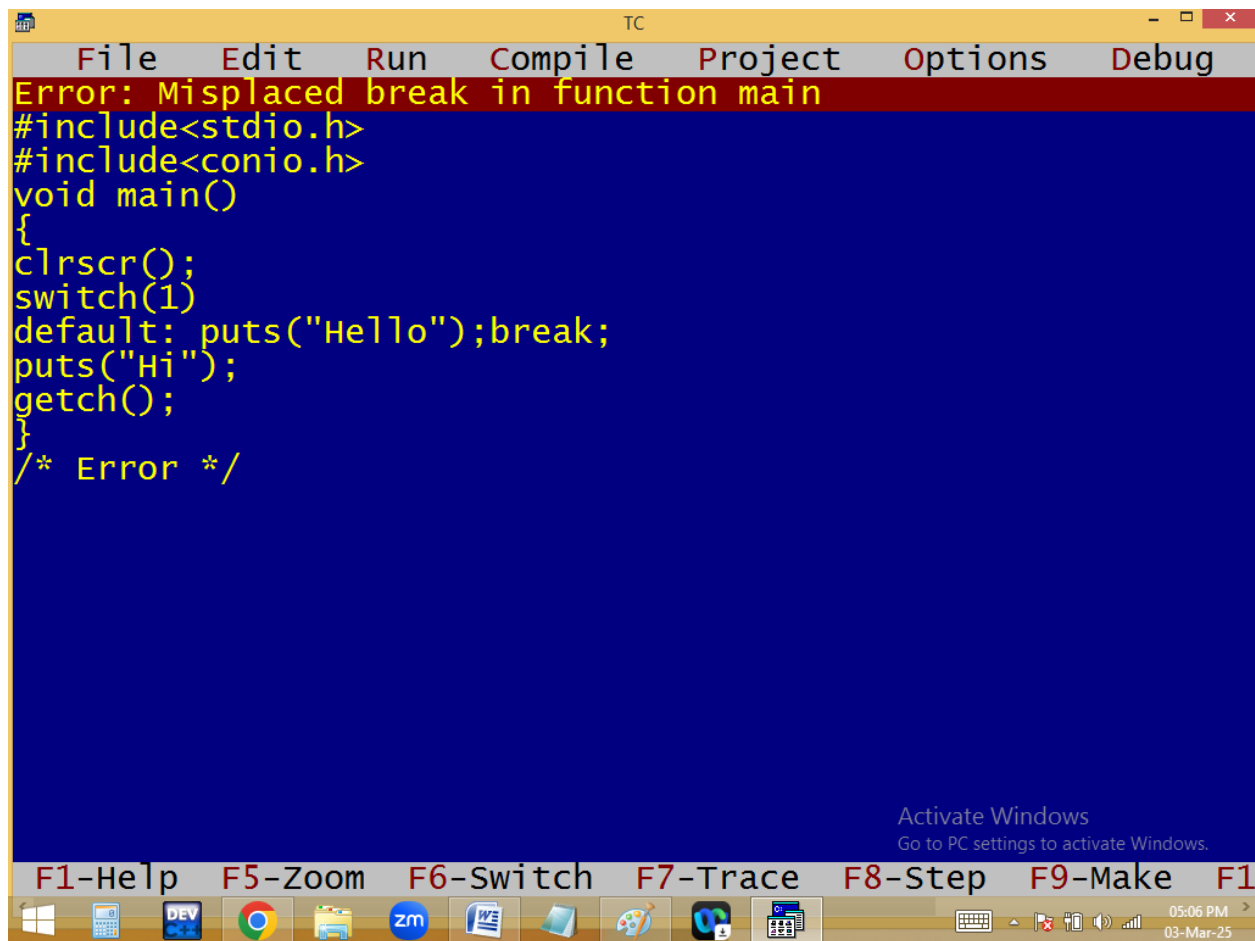


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 10 Col 16 Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
puts("Hi");
getch();
}
/* Blank screen */
```

At the bottom of the IDE window, there is a text prompt: "Activate Windows Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar. It features several icons: Windows Start button, Task View, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, a folder icon, a game controller icon, and a calendar icon. The taskbar also displays the system clock as "05:04 PM" and the date "03-Mar-25".

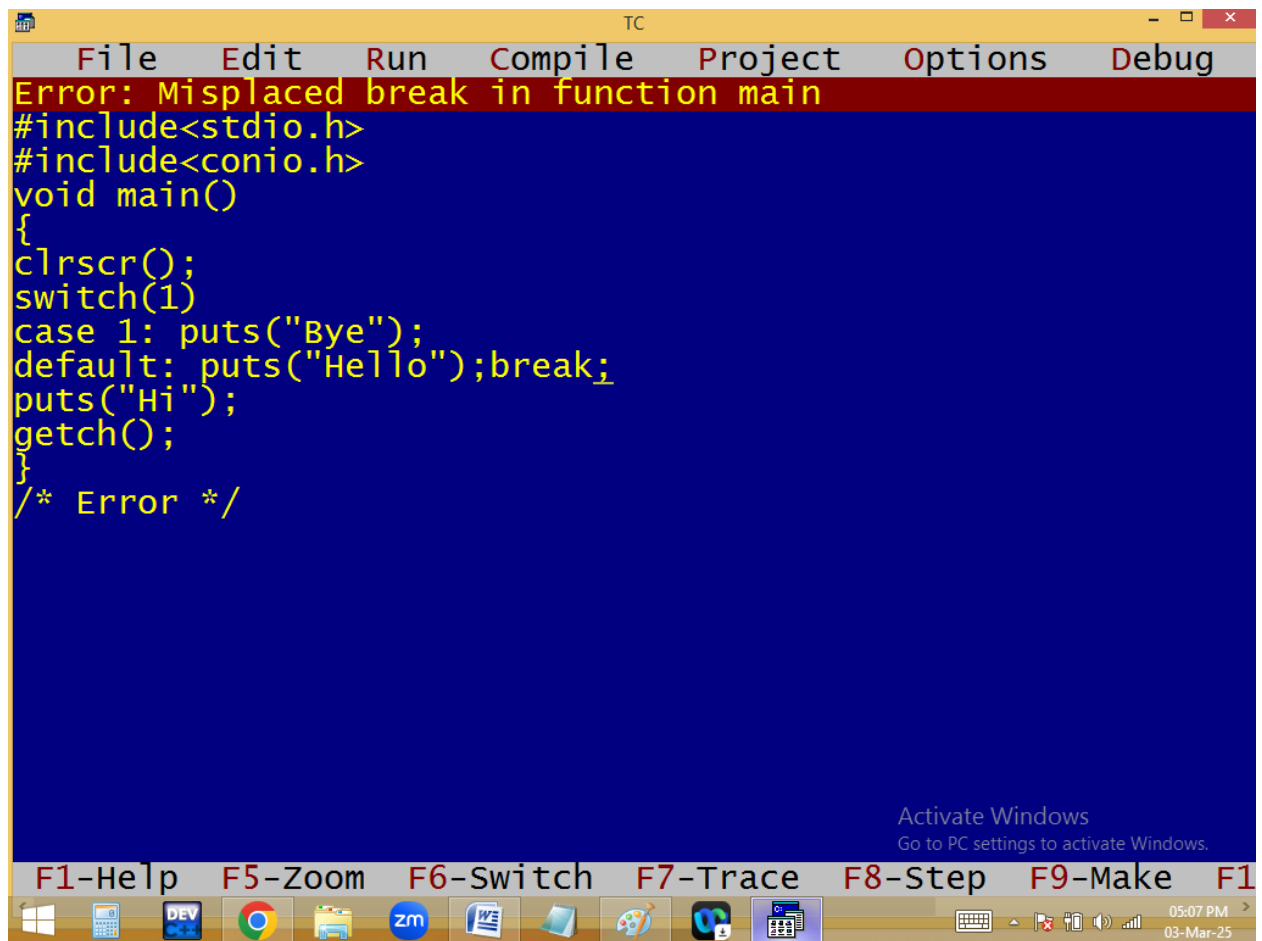
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run



The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the code editor displays the message "Error: Misplaced break in function main". The code editor has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
default: puts("Hello");break;
puts("Hi");
getch();
}
/* Error */
```

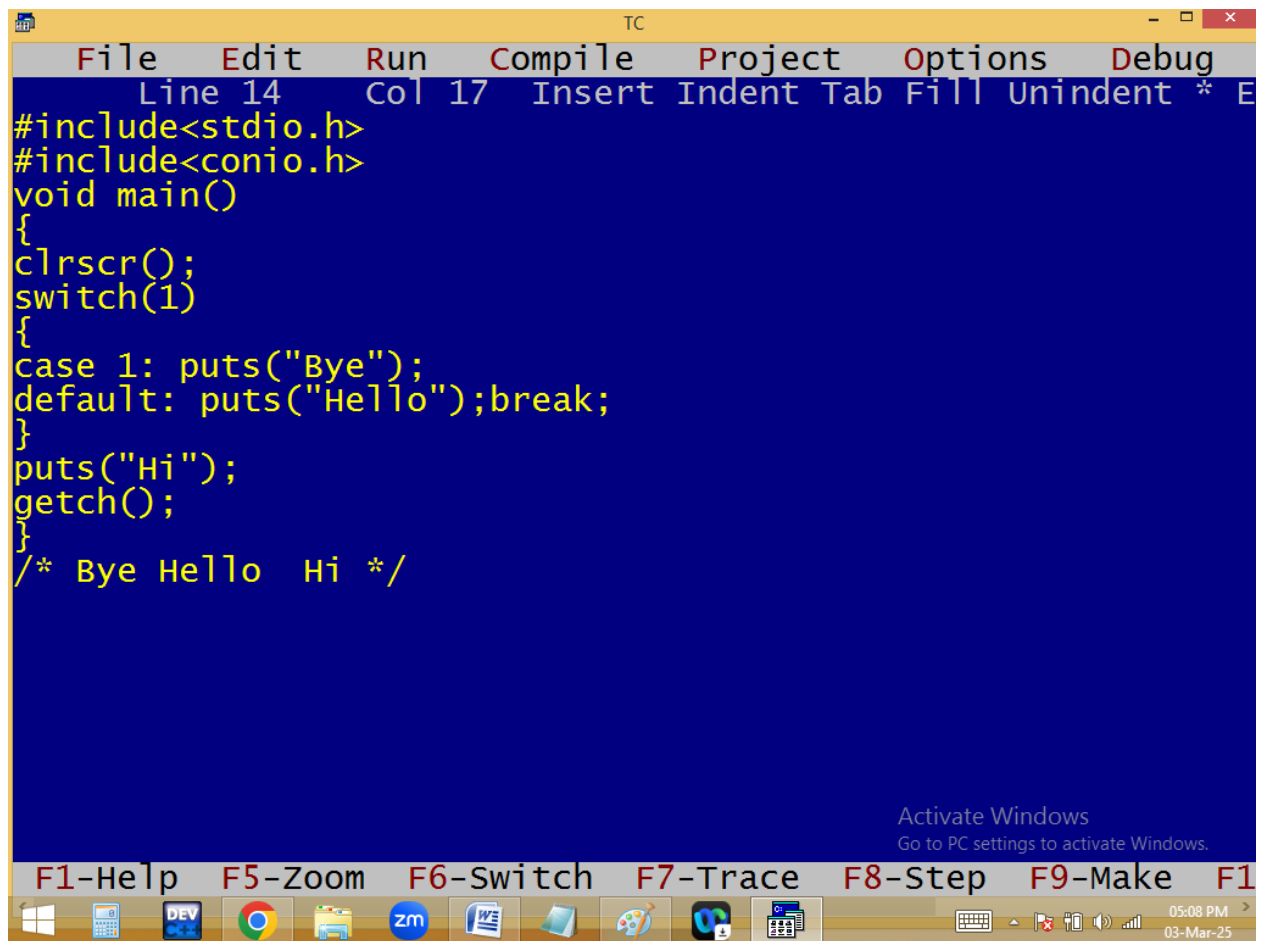
At the bottom of the code editor, there is a text prompt: "Activate Windows. Go to PC settings to activate Windows." Below the code editor is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, Task View, File Explorer, and several other applications. The system clock in the bottom right corner shows "05:06 PM" and "03-Mar-25".



The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the editor area states "Error: Misplaced break in function main". The code in the editor is as follows:

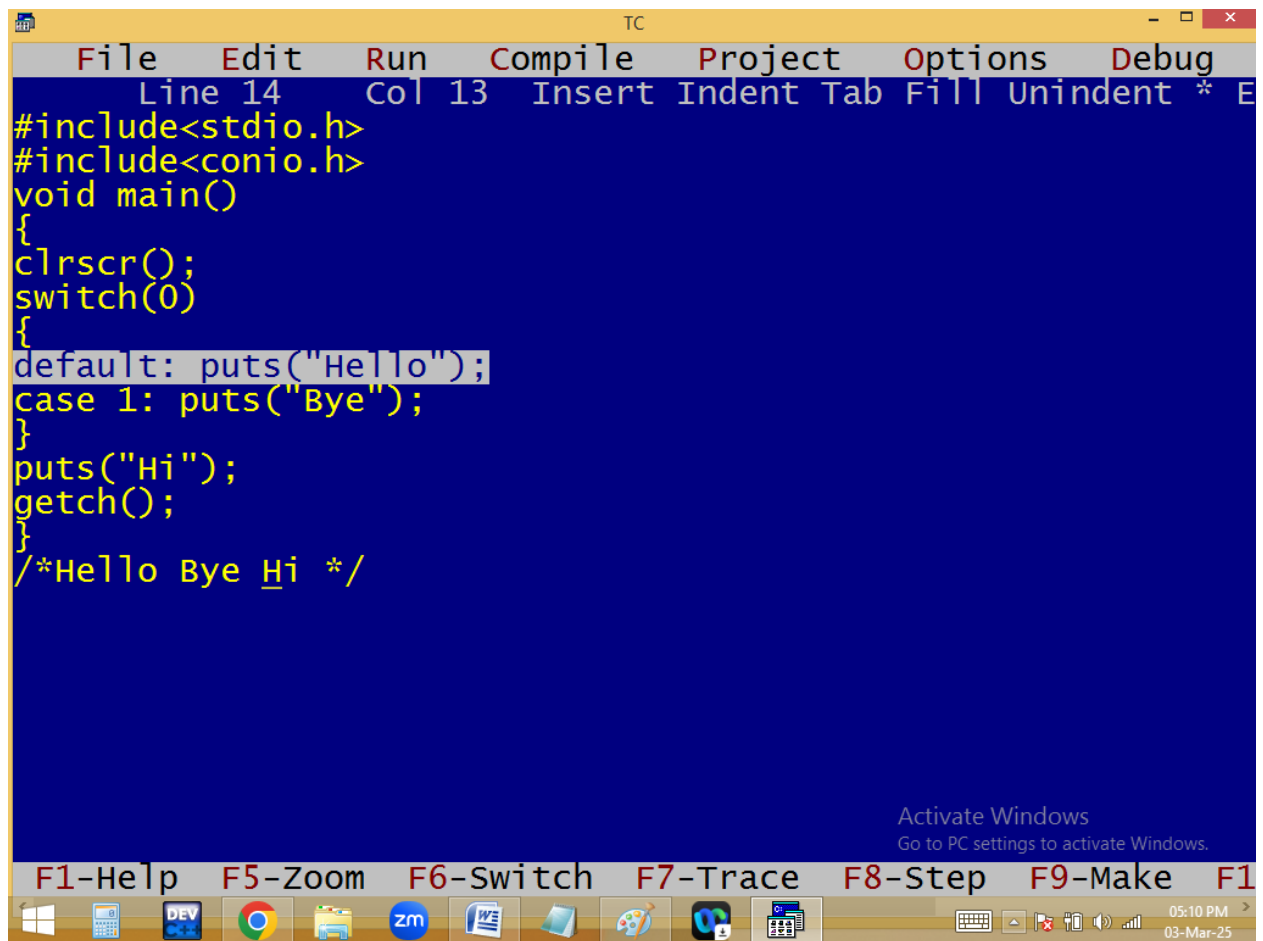
```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
case 1: puts("Bye");
default: puts("Hello");break;
puts("Hi");
getch();
}
/* Error */
```

At the bottom of the IDE window, there is a status bar with the text "Activate Windows" and "Go to PC settings to activate Windows." Below this, a row of function key shortcuts is displayed: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, Task View, File Explorer, and several other applications. The system clock in the bottom right corner shows "05:07 PM" and "03-Mar-25".



```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 17 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
{
case 1: puts("Bye");
default: puts("Hello");break;
}
puts("Hi");
getch();
}
/* Bye Hello Hi */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
05:08 PM
03-Mar-25
```



```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 13 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(0)
{
default: puts("Hello");
case 1: puts("Bye");
}
puts("Hi");
getch();
}
/*Hello Bye Hi */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
```


TC

File Edit Run Compile Project Options Debug

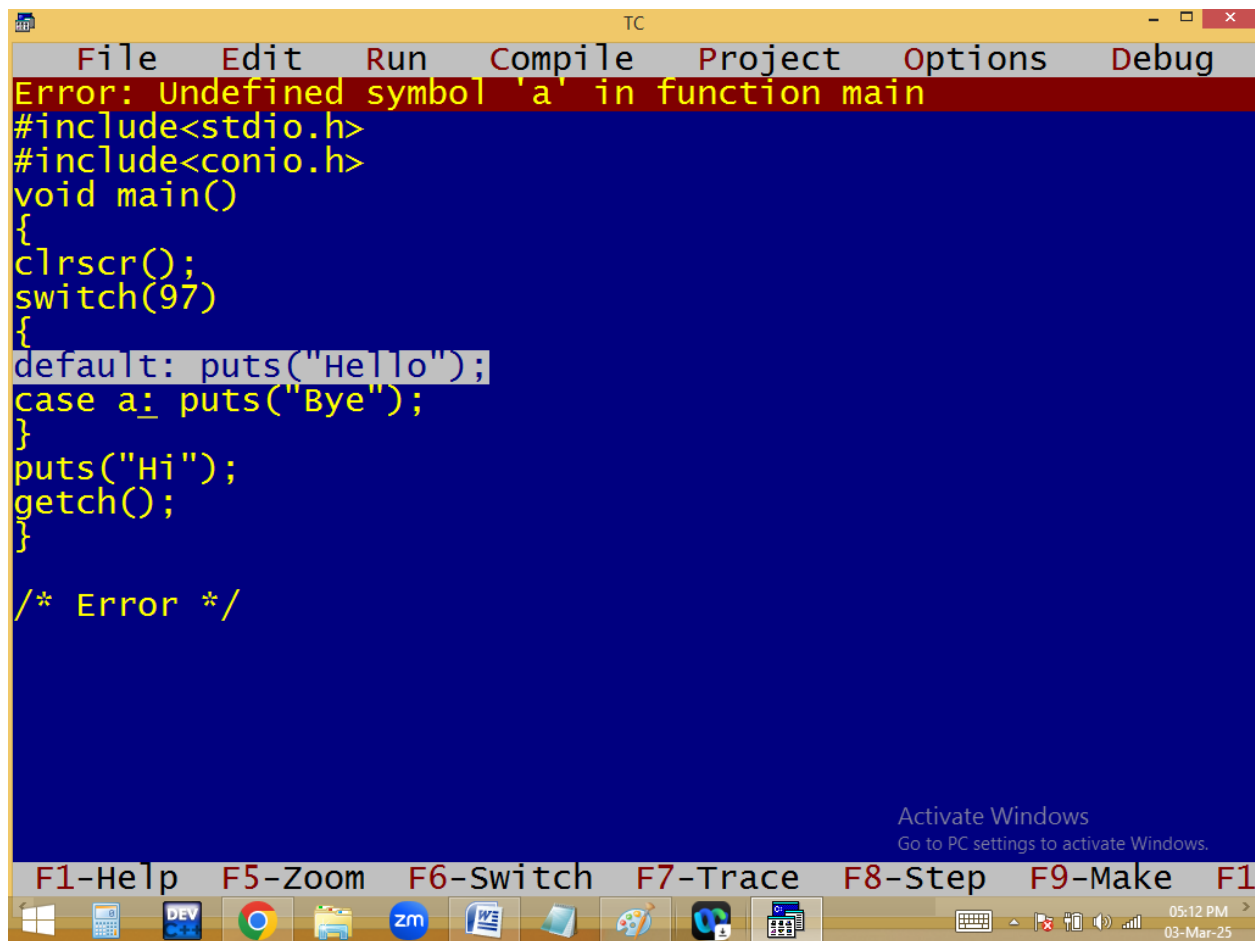
Error: Constant expression required in function main

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1.1)
{
default: puts("Hello");
case 1.1: puts("Bye");
}
puts("Hi");
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

05:10 PM
03-Mar-25



The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the editor window states: "Error: Undefined symbol 'a' in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    clrscr();
    switch(97)
    {
default: puts("Hello");
case a: puts("Bye");
    }
    puts("Hi");
    getch();
}

/* Error */
```

At the bottom of the IDE, there is a status bar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, Task View, File Explorer, and several other applications. The system clock in the bottom right corner shows "05:12 PM" and "03-Mar-25".

TC

File Edit Run Compile Project Options Debug

Error: Constant expression required in function main

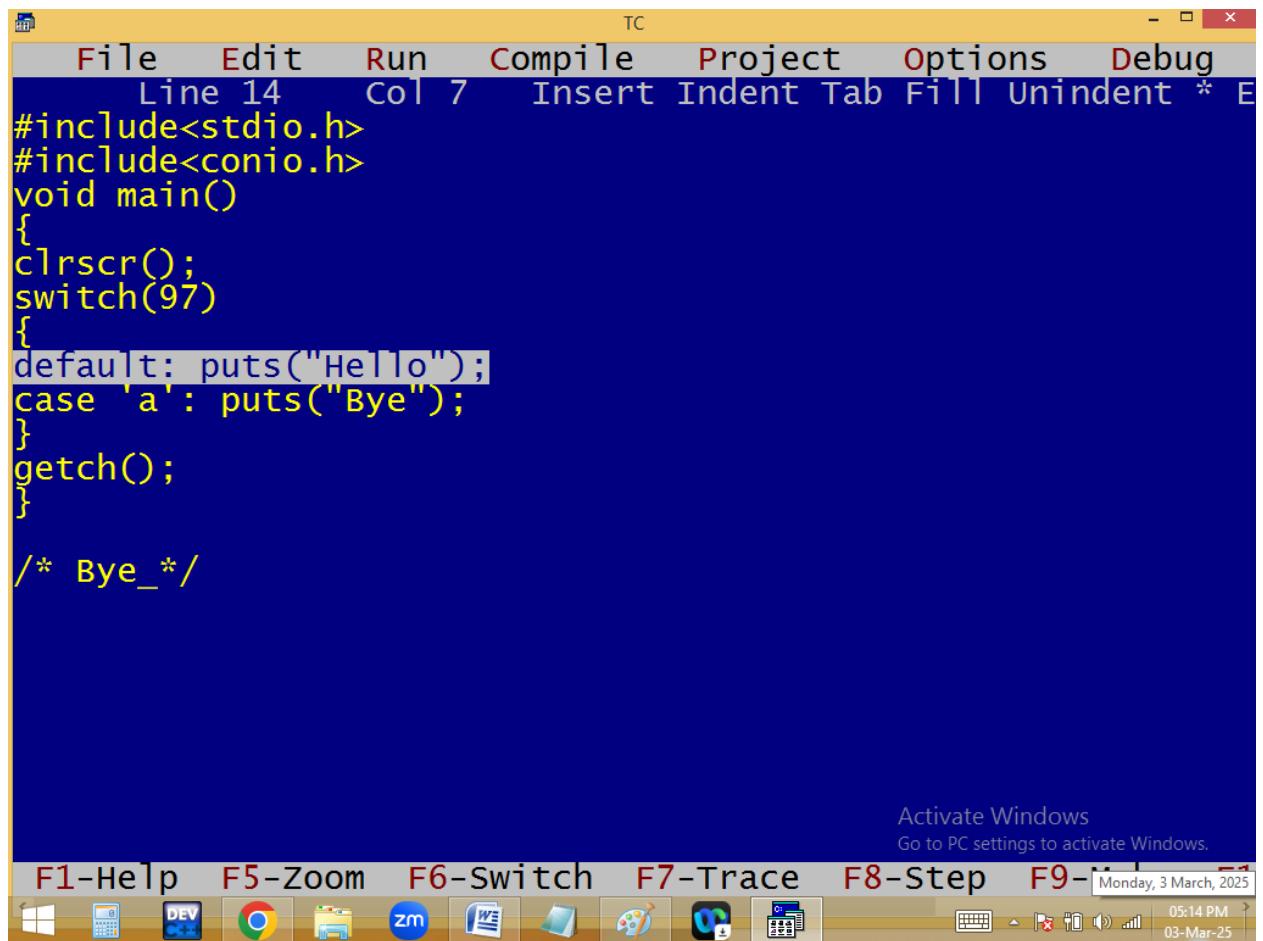
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=97;
clrscr();
switch(97)
{
default: puts("Hello");
case a: puts("Bye");
}
puts("Hi");
getch();
}

/* Error */
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

05:13 PM
03-Mar-25



```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 7 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(97)
{
default: puts("Hello");
case 'a': puts("Bye");
}
getch();
}

/* Bye_*/

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-
Monday, 3 March, 2025 05:14 PM 03-Mar-25
```

TC

File Edit Run Compile Project Options Debug

Error: Expression syntax in function main

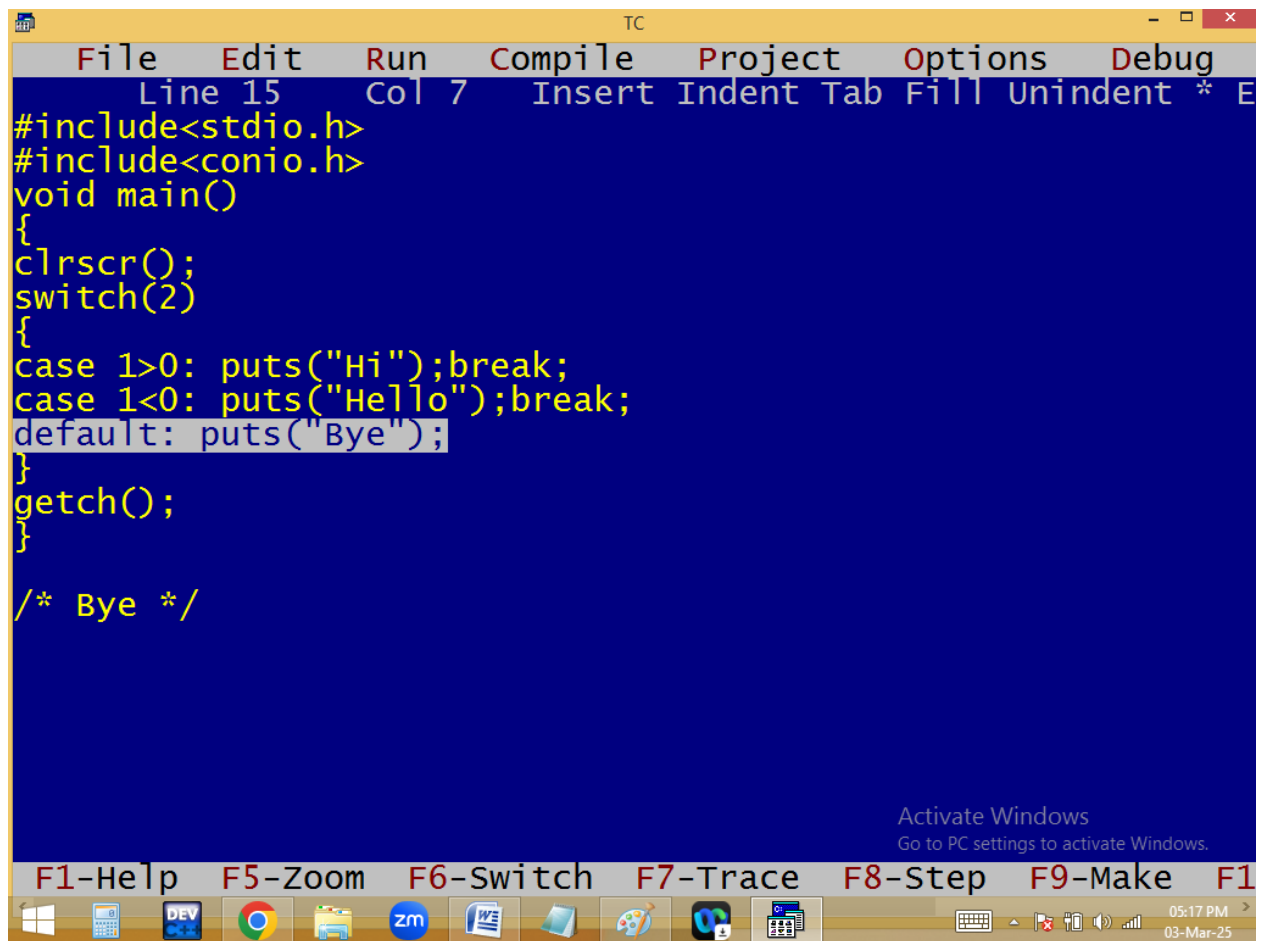
```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(2)
{
case >0: puts("Hi");break;
case <0: puts("Hello");break;
default: puts("Bye");
}
getch();
}

/* Error */
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

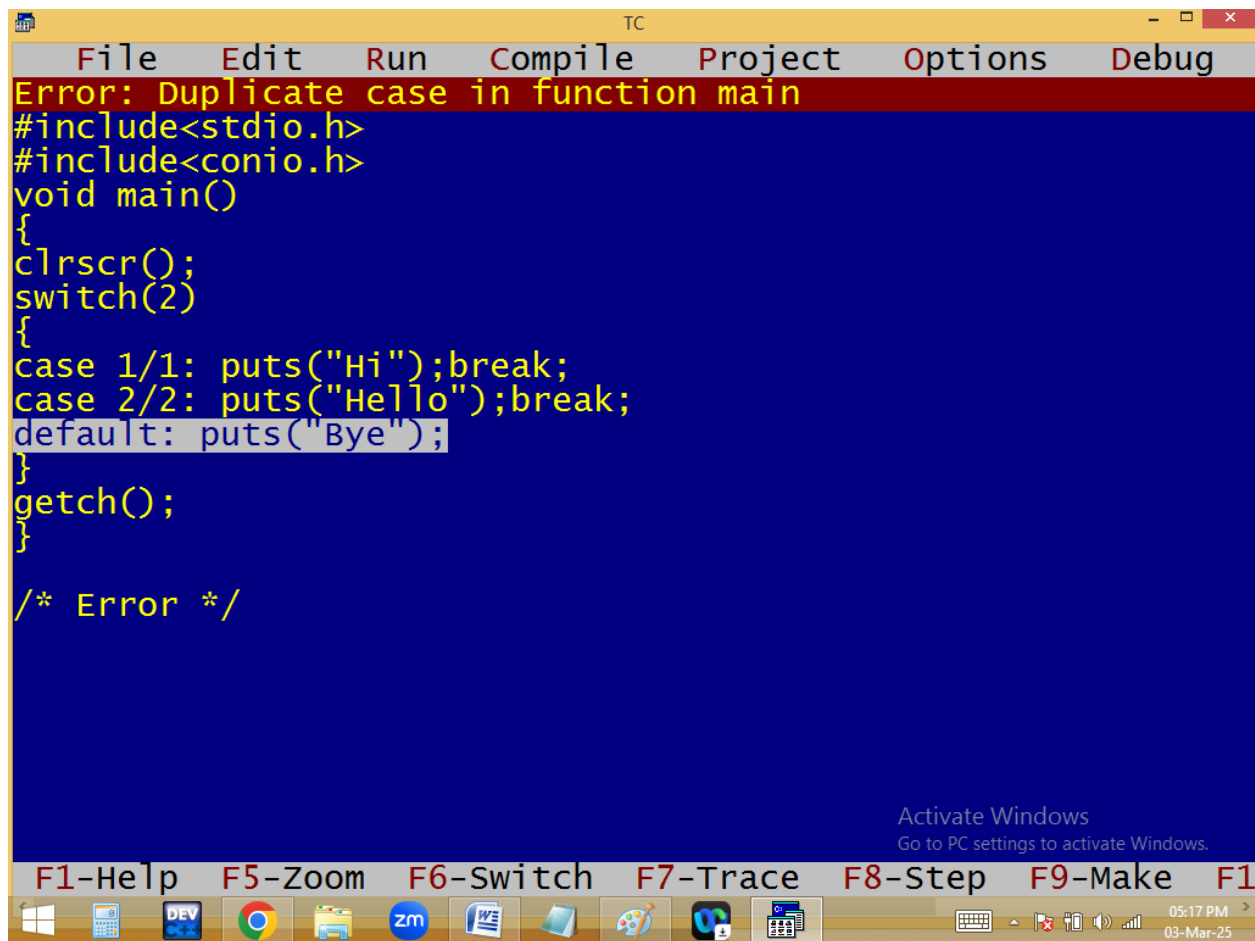
05:15 PM
03-Mar-25



```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 7 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(2)
{
case 1>0: puts("Hi");break;
case 1<0: puts("Hello");break;
default: puts("Bye");
}
getch();
}

/* Bye */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
05:17 PM
03-Mar-25
```

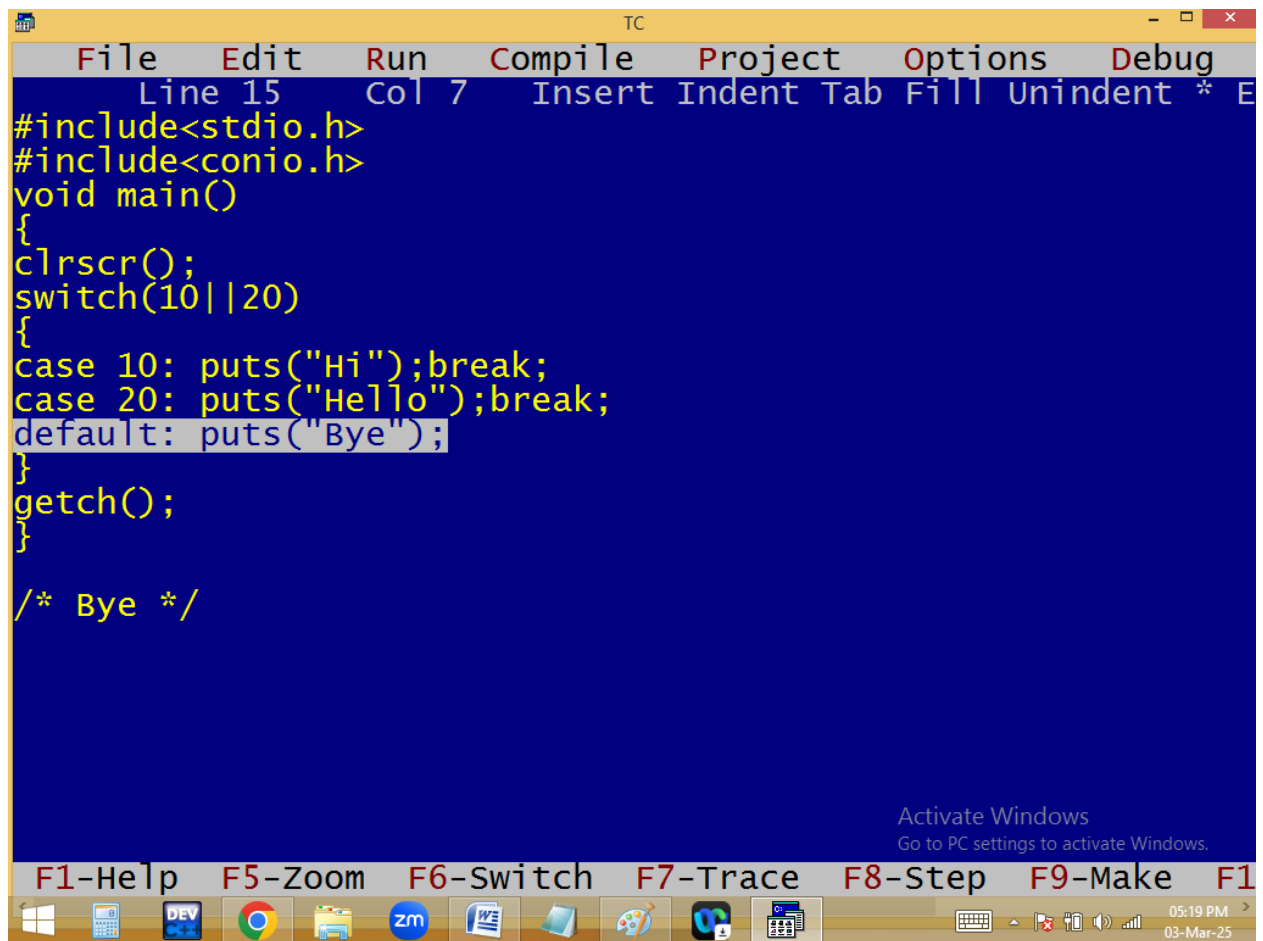


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top reads "Error: Duplicate case in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(2)
{
case 1/1: puts("Hi");break;
case 2/2: puts("Hello");break;
default: puts("Bye");
}
getch();
}

/* Error */
```

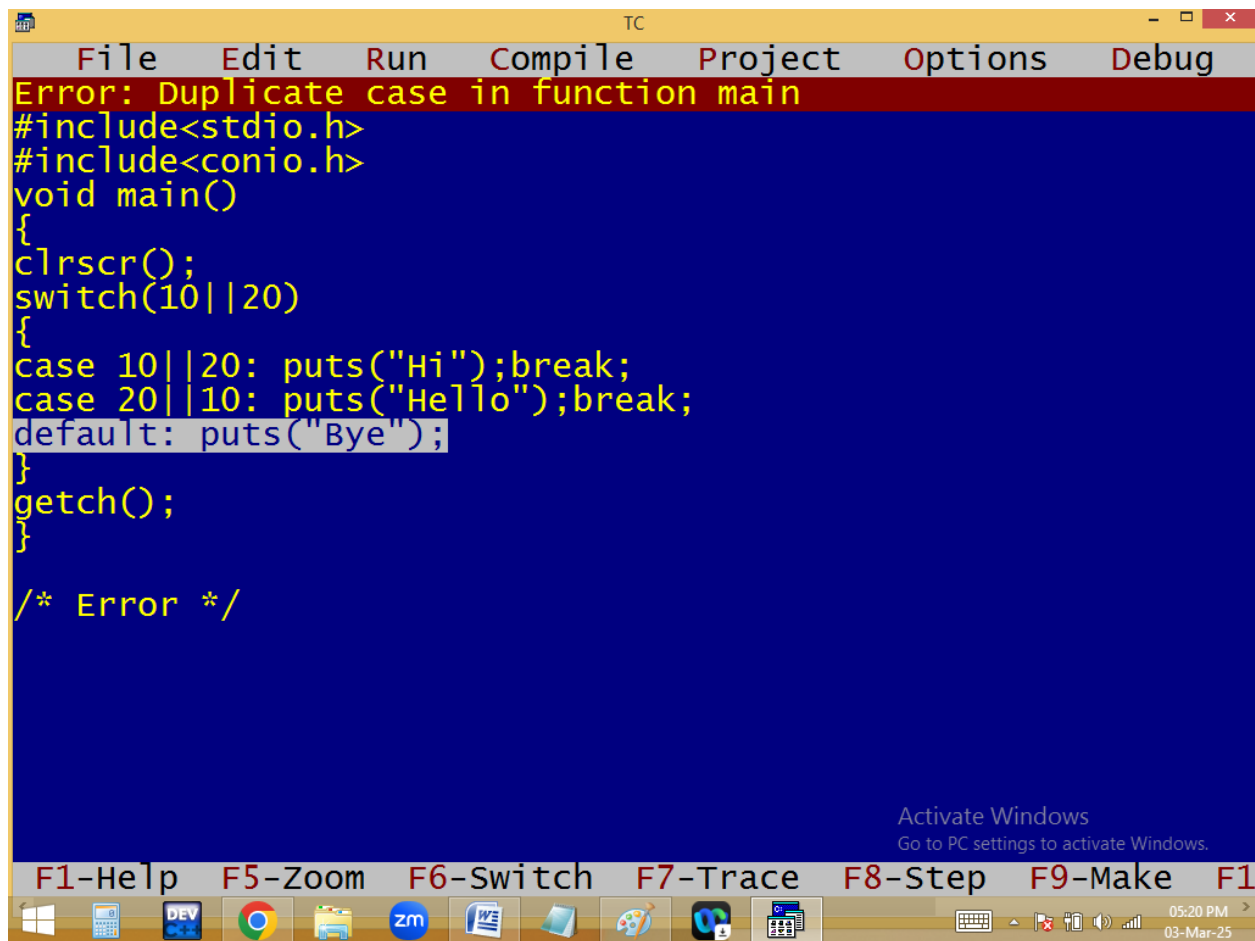
At the bottom of the IDE window, there is a status bar with the text "Activate Windows" and "Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar. It contains icons for the Start menu, Task View, File Explorer, Microsoft Edge, Zoom, Word, and several other applications. The system tray on the right shows the date and time as "05:17 PM" and "03-Mar-25".



```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 7 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(10||20)
{
case 10: puts("Hi");break;
case 20: puts("Hello");break;
default: puts("Bye");
}
getch();
}

/* Bye */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
05:19 PM
03-Mar-25
```

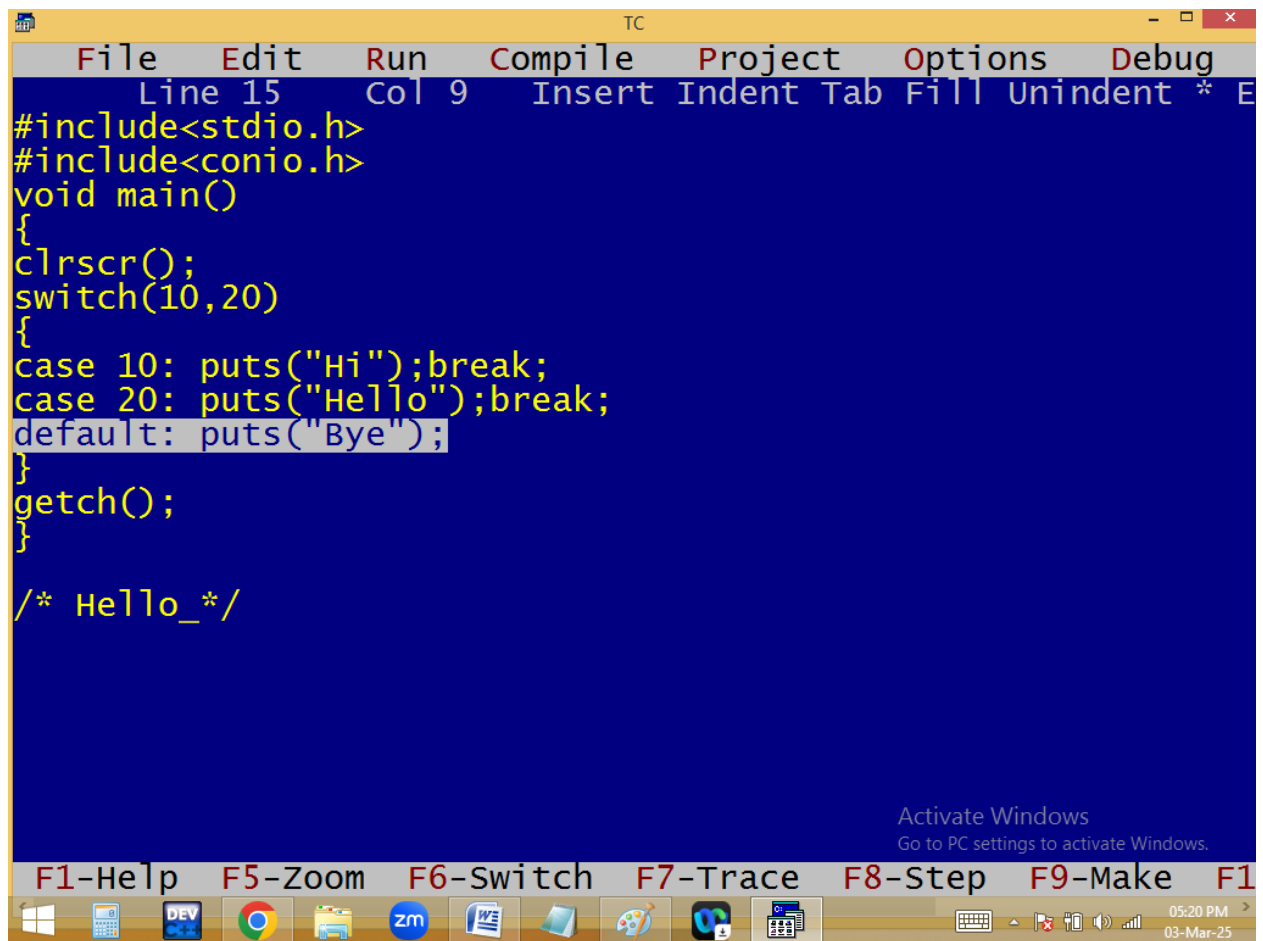
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top reads "Error: Duplicate case in function main". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(10||20)
{
case 10||20: puts("Hi");break;
case 20||10: puts("Hello");break;
default: puts("Bye");
}
getch();
}

/* Error */
```

The line "default: puts("Bye");" is highlighted with a light blue selection. At the bottom of the IDE window, there is a status bar with the text "Activate Windows" and "Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Microsoft Edge, Zoom, Word, and other applications. The system tray on the right shows the date and time as "05:20 PM" and "03-Mar-25".

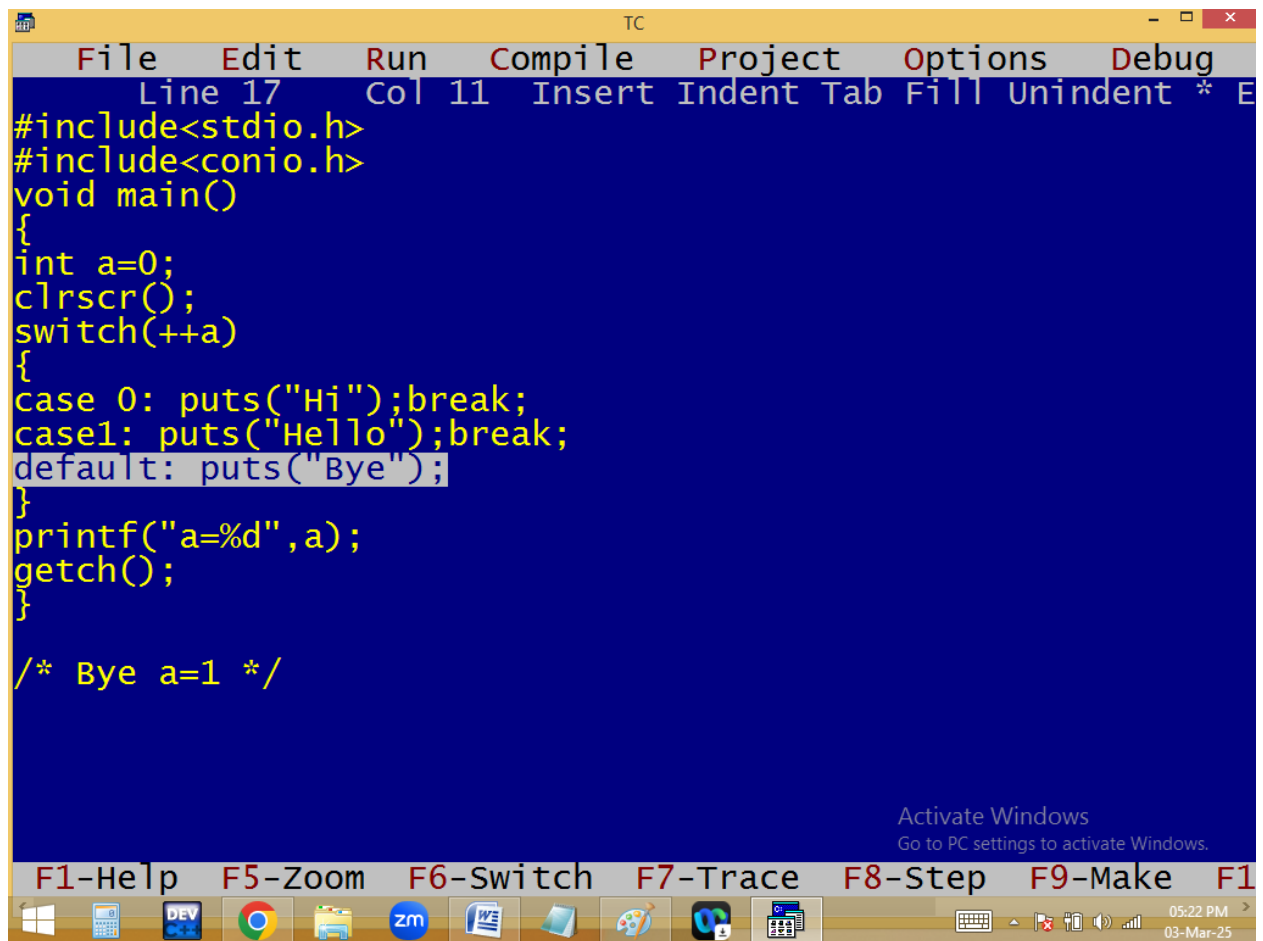
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run



```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 9 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(10,20)
{
case 10: puts("Hi");break;
case 20: puts("Hello");break;
default: puts("Bye");
}
getch();
}

/* Hello_*/

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
05:20 PM
03-Mar-25
```



```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(++a)
{
case 0: puts("Hi");break;
case1: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Bye a=1 */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
05:22 PM
03-Mar-25
```

TC

File Edit Run Compile Project Options Debug

Line 19 Col 20 Insert Indent Tab Fill Unindent * E

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
xyz:
switch(a++)
{
case 1: puts("Hi");break;
case 2: puts("Hello");break;
default: puts("Bye");
}
if(a<=2) goto xyz;
printf("a=%d",a);
getch();
}

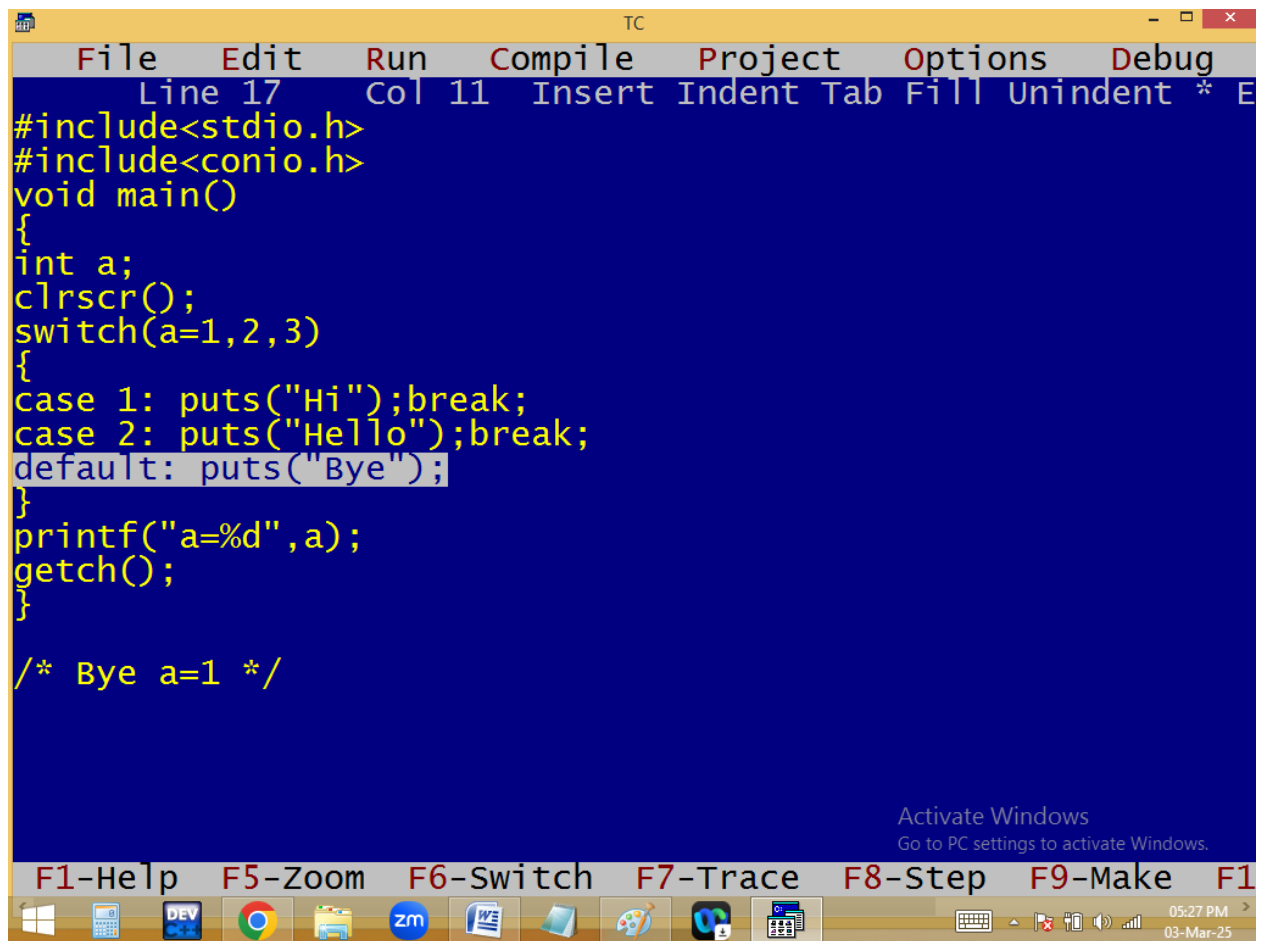
/* Bye Hi Hello a=3_*/
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

Webex

05:25 PM
03-Mar-25



```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a;
clrscr();
switch(a=1,2,3)
{
case 1: puts("Hi");break;
case 2: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Bye a=1 */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
```

switch(a = 1, 2, 3)

The diagram illustrates the execution of a switch statement. The text **switch(a = 1, 2, 3)** is shown. A red arrow points from the opening parenthesis to the first case, **a = 1**. Another red arrow points from the first case to the second case, **a = 2**. A third red arrow points from the second case to the third case, **a = 3**. The cases are listed vertically: **a = 1**, **a = 2**, and **a = 3**.

TC

File Edit Run Compile Project Options Debug

Line 18 Col 6 Insert Indent Tab Fill Unindent * E

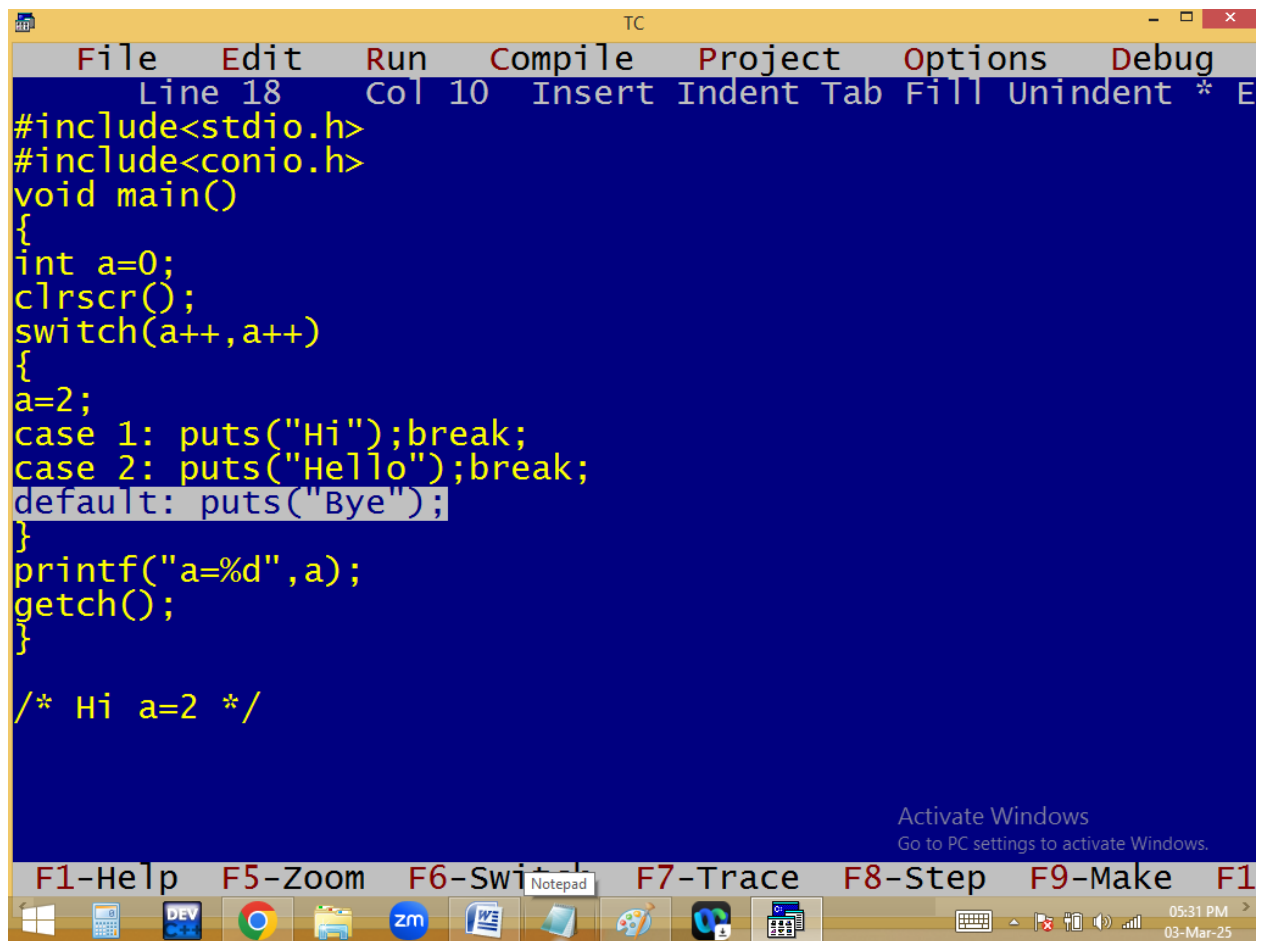
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a=1)
{
a=2;
case 1: puts("Hi");break;
case 2: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Hi a=1 */
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

05:29 PM
03-Mar-25

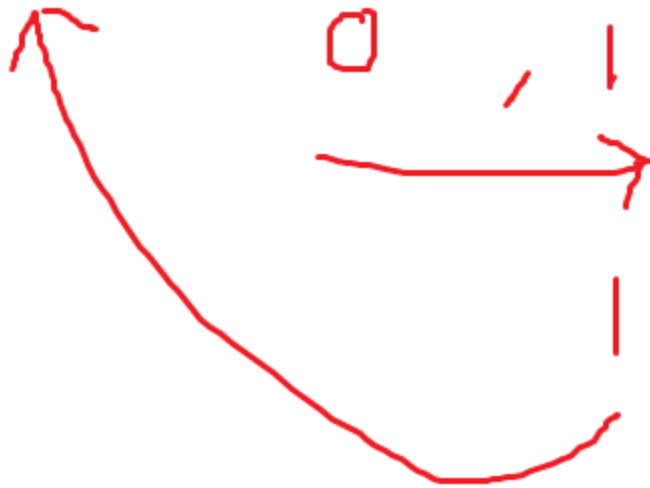


```
TC
File Edit Run Compile Project Options Debug
Line 18 Col 10 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a++,a++)
{
a=2;
case 1: puts("Hi");break;
case 2: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Hi a=2 */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
05:31 PM
03-Mar-25
```


a=0 ↗ 1 ↗ 2 ✓
switch(a++, a++)



```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a=a++)
{
case 1: puts("Hi");break;
case 2: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Bye a=0_*/
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

a=0;

switch(a=a++)

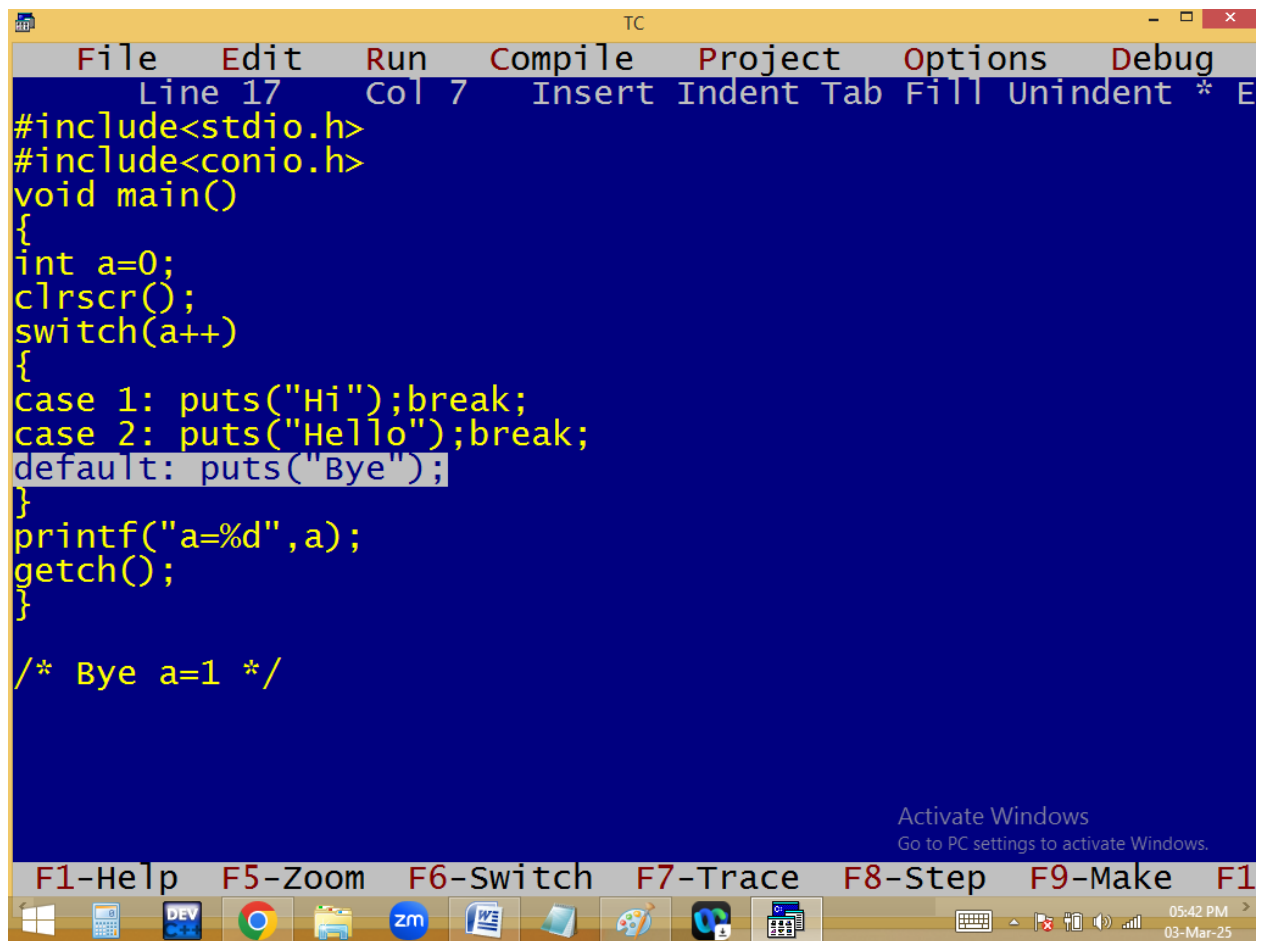
```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 10 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a=a++,++a)
{
case 1: puts("Hi");break;
case 2: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Hi a=1 */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
```

~~a=0~~ |

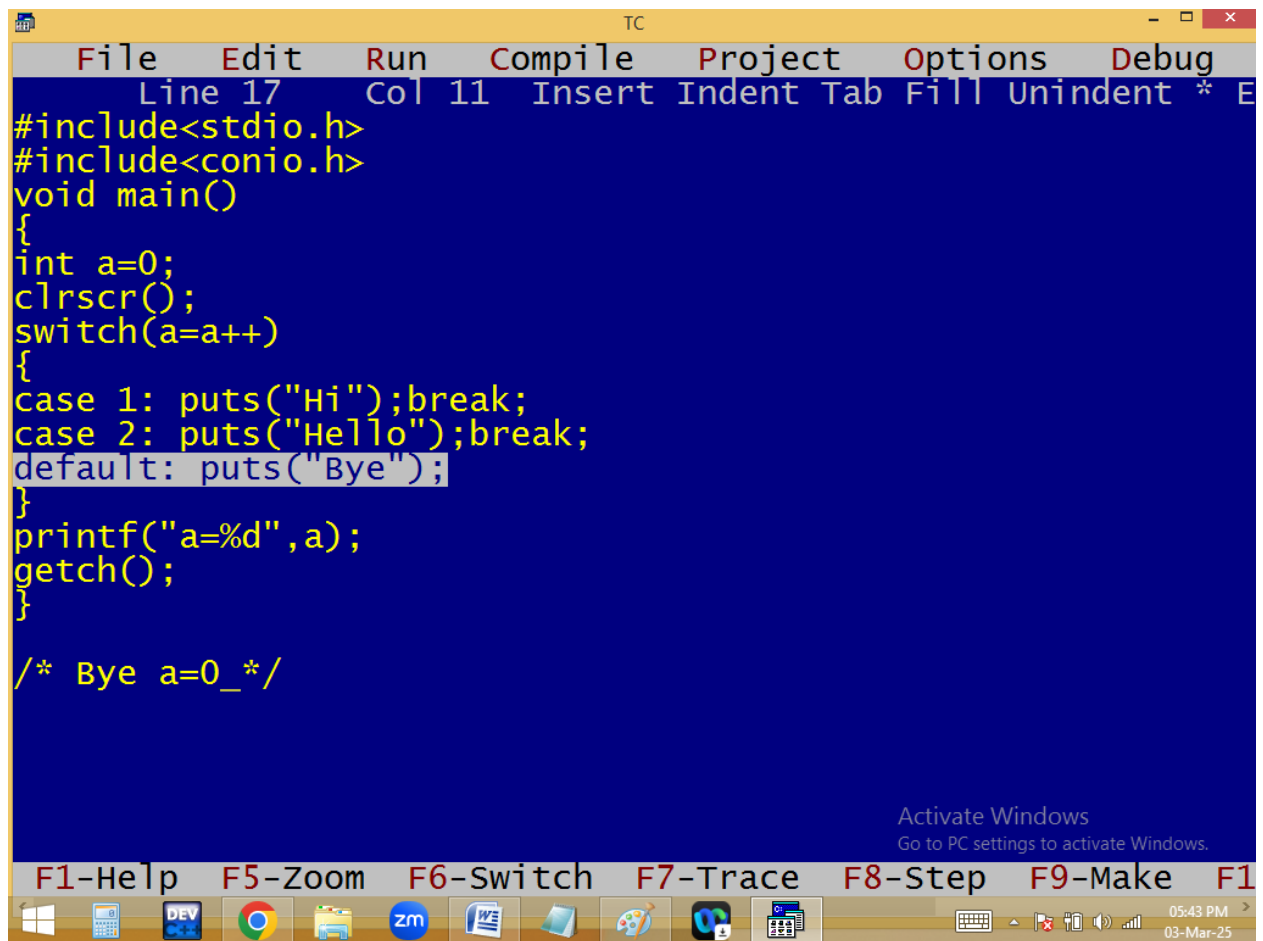
switch(a = ~~a++~~ , ++a)



```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 7 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a++)
{
case 1: puts("Hi");break;
case 2: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Bye a=1 */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
```



```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a=a++)
{
case 1: puts("Hi");break;
case 2: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Bye a=0_*/

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
```

```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 10 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a=a++,a++,a++)
{
case 1: puts("Hi");break;
case 2: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Hi a=2 */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
```

a=0

switch(a=a++, a++, a++)

a=0

switch(b=a++, a++, a++)

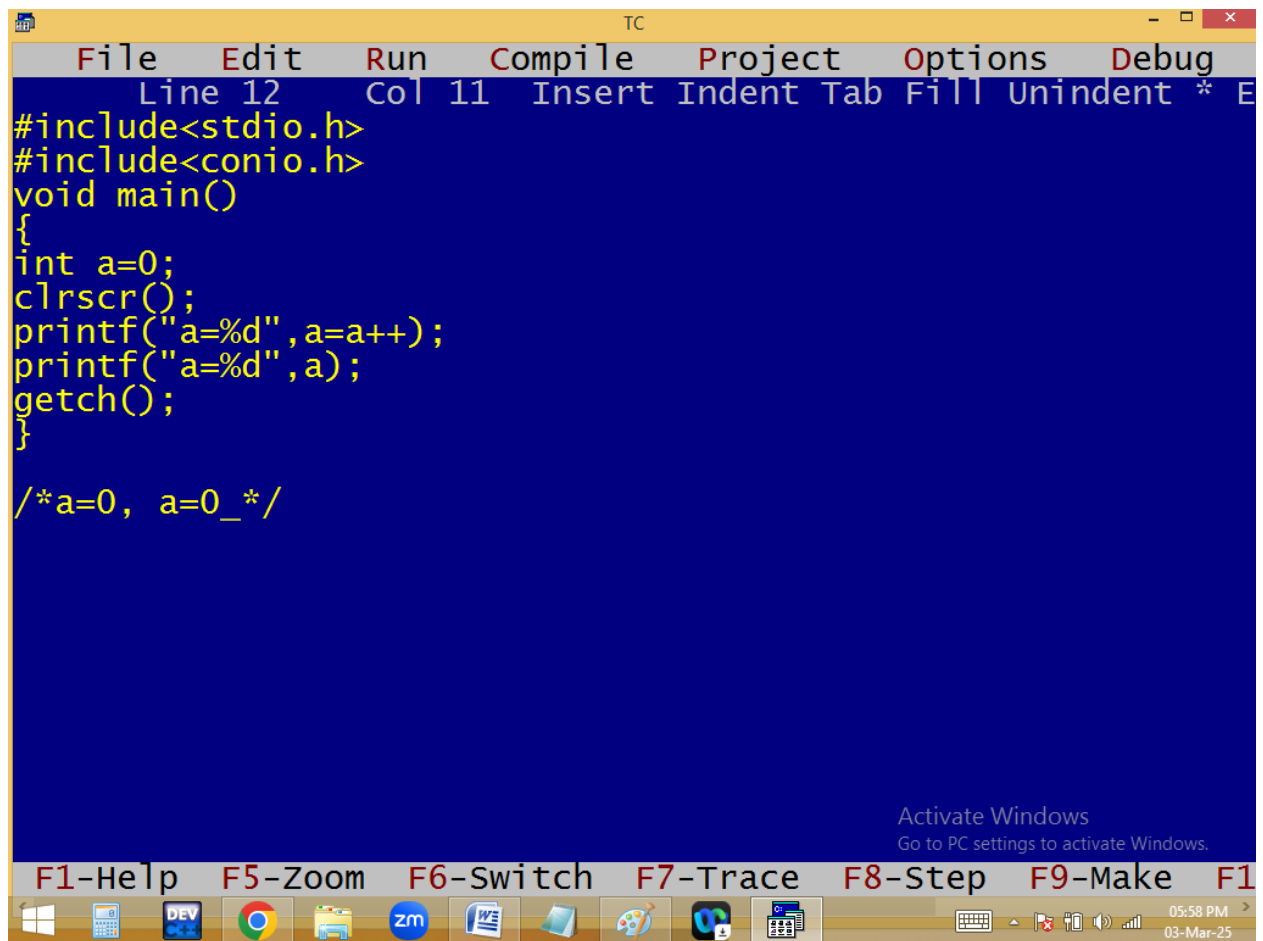
0

3

```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 6 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
a=a++;
printf("a=%d",a);
getch();
}

/*a=1 */

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit
```



```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
printf("a=%d",a=a++);
printf("a=%d",a);
getch();
}

/*a=0, a=0_*/

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
05:58 PM
03-Mar-25
```



```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
printf("a=%d",a++);
printf("a=%d",a);
getch();
}

/*a=0, a=1_*/

Activate Windows
Go to PC settings to activate Windows.
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
```